Stock Price Prediction using LSTM with Attention

PRESENTED BY

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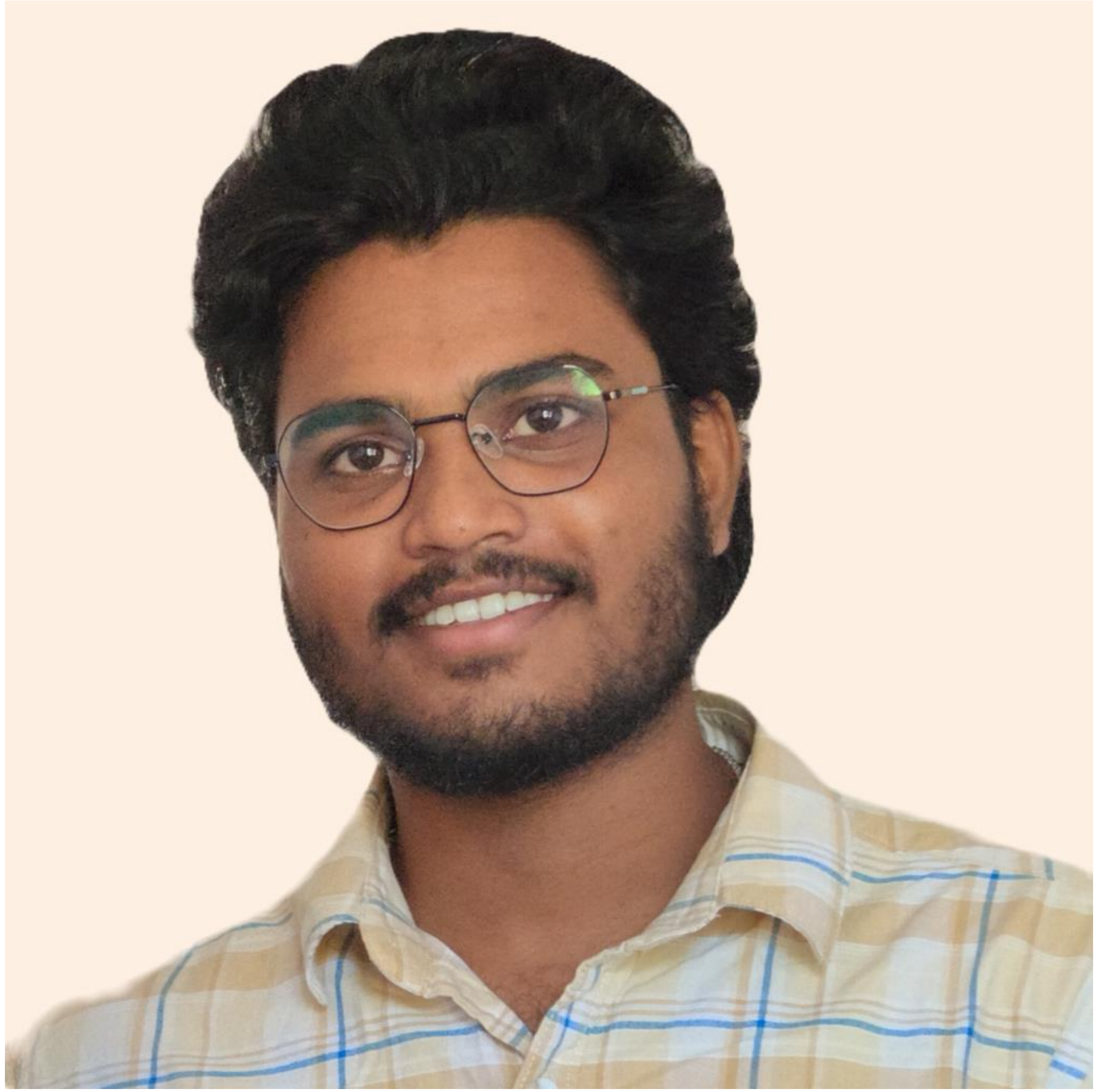


OUTLINE

- **Problem Statement** (Should not include solution)
- **Proposed System/Solution**
- **System Development Approach** (Technology Used)
- **Algorithm & Deployment**
- **Result (Output Image)**
- **Conclusion**
- **Future Scope**
- **References**

PROBLEM STATEMENT

- Predicting stock market trends is inherently complex due to its non-linear and volatile nature.

- Traditional statistical models often fail to capture long-term dependencies.

- Need for an intelligent model that can focus on relevant time steps for accurate predictions.

PROPOSED SOLUTION

- Use **Long Short-Term Memory (LSTM)** networks to capture sequential patterns.
- Integrate a **custom Attention Mechanism** to prioritize significant time steps.
- Train on historical stock data to predict future closing prices.

SYSTEM APPROACH

- **Data Collection**: Using Yahoo Finance API (yfinance)
- **Python**
- **TensorFlow/Keras**
- **Preprocessing**:
- Normalization using MinMaxScaler
- Sequence slicing (e.g., 60 previous days → 1 prediction)
- **Model Building**:
- LSTM layers
- Custom Attention layer
- Dense output layer

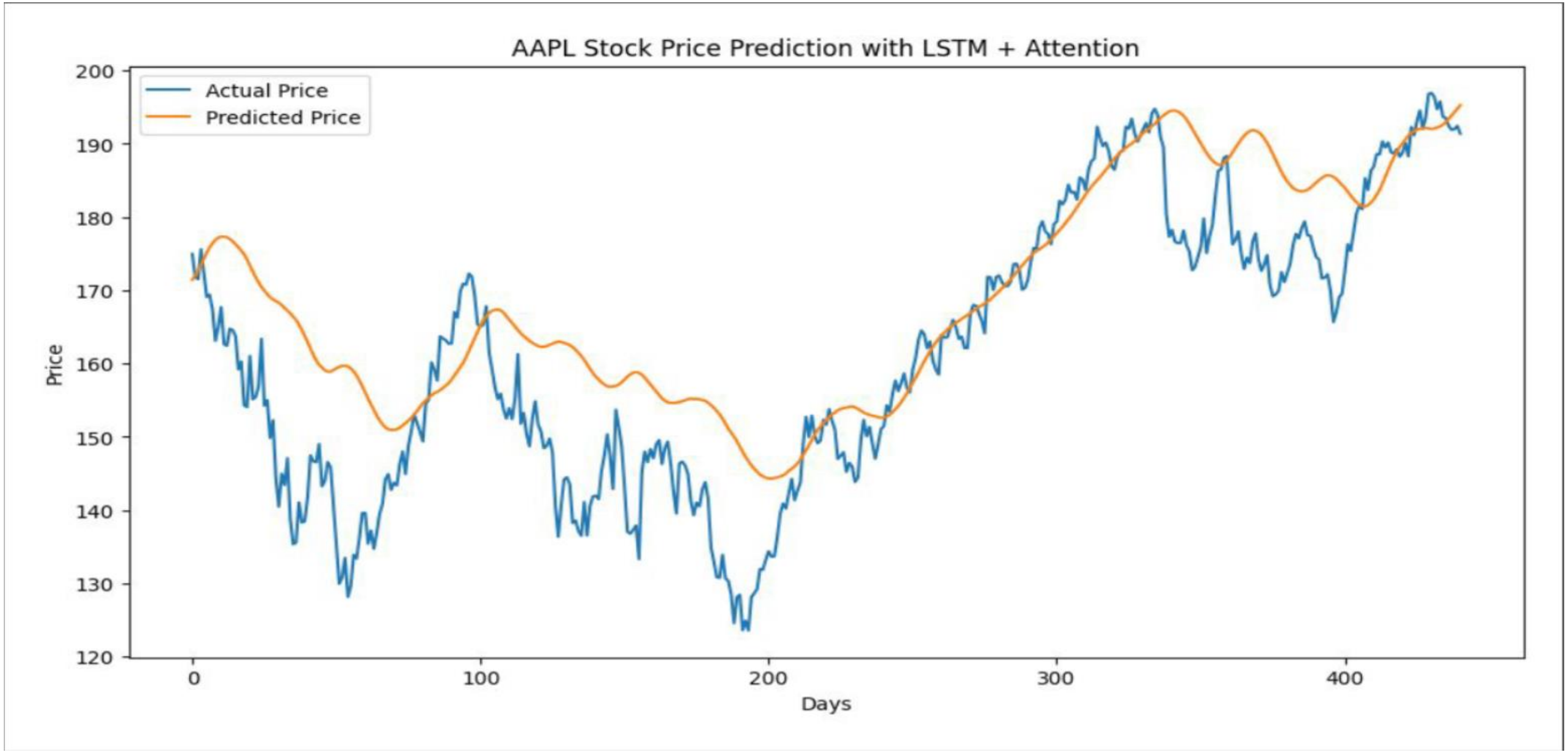
ALGORITHM & DEPLOYMENT

- **Algorithm Workflow:**
- Input historical price sequences
- Pass through LSTM layers
- Apply Attention mechanism
- Predict next-day price using dense layers
- **Deployment Ideas:**
- Flask or Streamlit web app for real-time predictions
- Hosted on platforms like Heroku or AWS EC2

RESULT

- Model shows improved accuracy compared to baseline LSTM
- Attention layer helps the model focus on important inputs
- Visual graphs indicate strong correlation between actual and predicted values

RESULT (OUTPUT IMAGE)



CONCLUSION

- LSTM with Attention offers a robust solution for stock price prediction
- Attention increases interpretability and model efficiency
- The hybrid architecture is promising for financial forecasting

FUTURE SCOPE

- Add features like:
- Trading volume
- Opening price
- Technical indicators
- Experiment with:
- Bidirectional LSTM
- Transformer models
- Create an end-user application with real-time data feeds

REFERENCES

- TensorFlow/Keras documentation
- Scholarly articles on LSTM and Attention mechanisms
- Github Link : https://github.com/Porallanagaraju13/
- Data Set :https://finance.yahoo.com/

Thank you